Signature Compression for SQIsign via Weil Pairing Recovery

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October 23, 2026

Abstract

The SQIsign v2.0 specification [SQI25a] transmits the full 2×2 change-of-basis matrix $\mathbf{M}_{\mathrm{chl}}$ in every signature. Dropping one entry and recovering it from the Weil pairing is a technique introduced by Basso, Maino and Pope for FESTA [BMP23] and applied to SQIsignHD by Dartois et al. [DLRW24]. The v2.0 specification neither implements it nor states the recovery formula for its two-dimensional response. We state and prove that formula from the specification's own definitions. The Weil pairing of the transmitted basis of the challenge curve, raised to the determinant, times the Weil pairing of the auxiliary basis, is a primitive fourth root of unity. Hence the determinant is known modulo $2^{e_{\mathrm{rsp}}-n_{\mathrm{bt}}+1}$, one bit short of the width at which the matrix is carried, and that bit is the only free one. We give an adaptive pivot for the signatures whose leading entry is even. With a one-bit hint, basis-hint recomputation and metadata packing, the level I signature becomes 129 bytes, 19 below the 148 of the specification. The construction is checked on the 300 NIST known-answer vectors at levels I, III and V. We explain why the reduced Tate pairing on the transmitted basis is useless here: its arguments are 2^{F-N}-th multiples of full-order points, so its values have order at most 2^{2N-F}. On the deterministic bases scaled to order 2^m a single Tate ladder recovers the determinant once $2m - F$ reaches the precision, on all 300 vectors. At level I that is one ladder of 187 steps against two of 128. The round-3 specification of September 2026 also transmits the full matrix; the same recovery at its parameters gives 176, 269 and 353 bytes against 200, 306 and 406, measured on its 300 known-answer vectors.

Keywords. SQIsign, isogeny-based cryptography, signature compression, Weil pairing, post-quantum cryptography.

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1 Introduction

SQIsign [DKLPW20] is an isogeny-based post-quantum digital signature scheme whose combined public-key-plus-signature size (213 bytes at NIST Level I) is roughly 7× smaller than FN-DSA and 17× smaller than ML-DSA. Cryptography Additional Signatures standardization process [NIST] in 2025, with an updated v2.0 specification [SQI25a] based on the SQIsign2D-West construction of Basso et al. [BDD+24]. The v2.0 scheme replaces the KLPT-based response of v1.0 with interpolation data transmitted as a 2×2 change-of-basis matrix $\mathbf{M}_{\text{chl}}$ over $\mathbb{Z}/2^N\mathbb{Z}$, yielding 148-byte signatures at Level I. A round-3 specification with new parameters followed in September 2026 [SQI26]; this paper treats SQIsign v2.0 and v2.0.1 (round 2), and round 3 in Section 8.

Compact signatures are especially valuable in bandwidth-constrained settings: TLS certificate chains, embedded attestation protocols, and blockchain transaction throughput all improve with smaller signatures. Every byte removed from the wire format compounds across millions of handshakes or blocks.

The determinant-recovery idea is known. Recovering a dropped matrix coefficient from the Weil pairing determinant relation was introduced in the context of FESTA by Basso, Maino, and Pope [BMP23]. They observed that "given three of the four coefficients representing the scaled points, it is possible to retrieve the fourth one via the compatibility of the Weil pairing with isogenies." Nakagawa and Onuki applied the same principle in QFESTA [NO24]. Dartois et al. deployed it in SQIsignHD [DLRW24], where the response matrix is compressed to three of four scalars with recovery via the determinant equation $a_1 b_2 - b_1 a_2 \equiv kq \pmod{\ell^f}$. The SQIsignHD paper further notes the invertibility pivot: "If a_1 is invertible, we can recover b_2 from the other scalars ... and we can recover a_2 otherwise." The earlier SIDH key-compression literature also uses pairing-based scalar recovery [CJL+17, ZSP+18, NR19].

The specification gap. Despite this lineage, the SQIsign v2.0 specification [SQI25a] transmits all four entries of $\mathbf{M}_{\text{chl}}$ (Table 3 of the spec). Pairings appear only in Algorithm 2.5 (ChangeOfBasis) to *compute* the full matrix during signing; no algorithm, footnote, or wire format in the specification implements the drop-and-recover compression for the dim-2 response structure. The v2.0.1 update [SQI25b] does not change this. Nakagawa and Onuki [NO25] confirm that SQIsign2DPush signatures "can be compressed similarly to SQIsign-v2.0," corroborating that the v2.0 baseline does not apply scalar recovery.

Lineage. We first wrote the technique down for the reference implementation's maintainers on 8 July 2026, as issue 19 of its repository, "Technical notes on signature compression techniques" [Ray26]. Luca De Feo replied for the team the same day that "we may introduce some compression in round 3, depending on performance". On 7 October 2026 he added that "we will give a closer look at compression for the next round". The issue cites a comment in `biextension.c` of the round-2 reference at tag `nist-v2` (commit `91e9e46`) [SQI25c]. It reads: "If we work with full order points, it would be faster to use the Tate pairings rather than the Weil pairings; this is not implemented yet."

Contributions. We fill this gap with the following concrete results, each situated against the prior work it builds on.

(1) **The recovery formula for the v2.0 response matrix, with a proof from the specification** (§3). Let (A, B) be the deterministic basis of $E_{\text{chl}}[2^N]$ and (C, D) that of $E_{\text{aux}}[2^{N-r_{\text{rsp}}}]$

as the verifier scales them, $\omega = e_{2^N}(A, B)$ and $\eta = e_{2^{N-r_{\mathrm{rsp}}}}(C, D)$. Theorem 3.2 states that $\omega^{\det \mathbf{M}}\eta$ is a primitive fourth root of unity for every honestly generated signature, so

$$\det(\mathbf{M}) \;\equiv\; \log_\omega\!\left(\eta^{-1}\right) \pmod{2^g}, \qquad g = e_{\mathrm{rsp}} - n_{\mathrm{bt}}.$$

The proof uses only bilinearity, alternation, compatibility with isogenies and the signer's Kani square. The response degree q cancels against the auxiliary degree $2^{e'} - q$. The formula is confirmed on all 300 NIST KAT vectors.

(2) **Precision: one free bit** (§3). The matrix is carried modulo 2^N with $N = g + 2$, the 2 being the extra torsion of the specification's $(2, 2)$-isogeny chain. The theorem fixes the parity of the quotient $(\det \mathbf{M} - \log_\omega \eta^{-1})/2^g$, so a discrete logarithm taken at order 2^N determines $\det \mathbf{M}$ modulo 2^{N-1} and leaves a single free bit, bit $N - 1$.

(3) **Adaptive pivot** (§4). The determinant has 2-adic valuation r_{rsp}. When it is odd and M_{00} is even, M_{01} is odd and the decompressor drops M_{10} instead of M_{11}, with the path read off the parity of M_{00}. We report how often each case occurs on the KAT vectors.

(4) **Compression from 148 to 129 bytes** (§5): the Weil recovery (148 to 133), the one-bit hint (133 to 132), basis-hint recomputation (132 to 130) and metadata packing (130 to 129), with the bit layouts.

(5) **Pairing selection and negative results** (§6). The reduced Tate pairing on the transmitted basis is degenerate because its arguments are 2^{F-N}-th multiples of full-order points, not because of the final exponentiation. We derive the order of its values as a function of the ladder length and measure it on every order from 2^N to 2^F. On the full-order points a single Tate ladder recovers the determinant, because the reduced Tate pairing is alternating on these curves; we report where that becomes cheaper than two Weil ladders. We also document why the hint cannot be dropped by signer retry and why 128 bytes needs a signer-side canonical form.

(6) **Implementation, costs, and round 3** (§7, §8). A pure Rust `no_std` verifier validates all three formats on the 300 KAT vectors; we break the cost of compressed verification into its components. On the round-3 parameters the same recovery gives 176, 269 and 353 bytes, measured on the 300 round-3 KAT vectors.

1.1 Related work

The isogeny key-compression literature begins with Costello et al. [CJL+17], who introduced point compression for SIDH public keys. Zanon et al. [ZSP+18] (journal version [ZSP+19]) accelerated SIDH key compression via Tate pairings and reverse basis decomposition. Naehrig and Renes [NR19] developed dual-isogeny-based compression for SIDH. Pereira and Barreto [PB21] eliminated the pairing entirely via Pohlig-Hellman on the elliptic curve group; we discuss the applicability of this technique to SQIsign decompression in §9.

For SQIsign optimization, Aardal et al. [AAA+25] achieved record verification times (8.55 Mcycles on Raptor Lake, 5.6 Mcycles for uncompressed signatures) using assembly-level optimization and multi-core parallelism. introduced extension-field signing for faster verification. Kim et al. [KLY26] reduced quaternion precision budgets by 74 to 75%. De Feo et al. [DJWY26] provided the first vectorized (NEON) SQIsign implementation.

Robert [Rob24] and Pope et al. [PRRS+25] developed faster cubical/biextension pairing algorithms with direct applicability to SQIsign; our Weil pairing computation follows their approach.

A complete security proof for SQIsign was given by Aardal et al. [ABDPW25].

1.2 Notation

The specification writes f for the 2-adic valuation of $p+1$ (Table 3 and Section 4.2.1 of [SQI25a]). We write F for it, to keep f free, and $v_2(n)$ for the 2-adic valuation of an integer n, the largest k with $2^k \mid n$. All other quantities are the specification's. Per level: $\lambda = 128, 192, 256$; $c = 5, 65, 27$; $F = 248, 376, 500$; $e_{\mathrm{rsp}} = 126, 192, 253$; $e_{\mathrm{chl}} = 122, 184, 247$.

Symbol	Meaning
λ	Security parameter, 128, 192 or 256.
p	The prime, $p = c \cdot 2^F - 1$ with c odd; Chapter 5 of [SQI25a].
F	$v_2(p+1)$, the exponent of the accessible 2-torsion (f in the specification); 248, 376, 500.
e_{rsp}	$\lceil \log_2 \sqrt{p} \rceil$, the response bound (Section 4.2.1); 126, 192, 253.
e_{chl}	$F - e_{\mathrm{rsp}}$, the bit length of the challenge; 122, 184, 247.
n_{bt}, r_{rsp}	Backtracking steps and $v_2(\deg \varphi_{\mathrm{rsp}})$, carried in the signature (Section 4.4.3).
e'	$e_{\mathrm{rsp}} - r_{\mathrm{rsp}} - n_{\mathrm{bt}}$, the length of the $(2,2)$-chain (Algorithm 4.2, line 20).
N	$e' + r_{\mathrm{rsp}} + 2 = e_{\mathrm{rsp}} - n_{\mathrm{bt}} + 2$, the bit width of a matrix entry (Algorithm 4.9, line 5).
g	$N - 2 = e_{\mathrm{rsp}} - n_{\mathrm{bt}}$, the precision of the recovery.
E_0	The curve $y^2 = x^3 + x$ with its fixed basis $B_0 = (P_0, Q_0)$ of $E_0[2^F]$.
$E_{\mathrm{pk}}, E_{\mathrm{com}}, E_{\mathrm{chl}}, E_{\mathrm{aux}}$	Public-key, commitment, challenge and auxiliary curves.
(A, B)	The deterministic basis of $E_{\mathrm{chl}}[2^F]$ scaled to order 2^N (Algorithm 4.9, lines 12 and 13).
(C, D)	The deterministic basis of $E_{\mathrm{aux}}[2^F]$ scaled to order $2^{e'+2}$ (lines 10 and 11).
$\mathbf{M}$	$\mathbf{M}_{\mathrm{chl}}$, entries $M_{00}, M_{01}, M_{10}, M_{11}$ in wire order, in $[0, 2^N)$.
$e_n(\cdot,\cdot)$	The Weil pairing of order n; μ_n the n-th roots of unity in $\mathbb{F}_{p^2}^*$.
ω, η	$e_{2^N}(A, B)$ and $e_{2^{e'+2}}(C, D)$.

2 Preliminaries

2.1 Supersingular elliptic curves and isogenies

We recall the definitions needed to fix notation. For a full treatment of elliptic curves we refer to Silverman [Sil86], and for a survey of isogeny-based cryptography to De Feo [DeF17].

Let p be a prime with $p \equiv 3 \pmod 4$. A supersingular elliptic curve over $\mathbb{F}_{p^2}$ is an elliptic curve whose $\mathbb{F}_{p^2}$-rational endomorphism ring is an order in a quaternion algebra rather than an imaginary quadratic field. SQIsign works exclusively with supersingular curves in Montgomery form $E_A : By^2 = x^3 + Ax^2 + x$ over $\mathbb{F}_{p^2}$, where the j-invariant is determined by A alone. The base curve is $E_0 : y^2 = x^3 + x$ (i.e., $A = 0$, $B = 1$) with known endomorphism ring $\mathcal{O}_0 \cong \mathrm{End}(E_0)$.

An isogeny $\varphi : E \to E'$ is a non-constant morphism of elliptic curves preserving the identity. Its degree $\deg(\varphi)$ equals the size of its kernel $\ker(\varphi)$. Every isogeny of degree d has a dual isogeny $\hat{\varphi} : E' \to E$ satisfying $\hat{\varphi} \circ \varphi = [d]_E$ and $\varphi \circ \hat{\varphi} = [d]_{E'}$, where $[d]$ denotes the multiplication-by-d map. An isogeny is cyclic if its kernel is a cyclic group, in which case it is determined up to post-composition by an automorphism by a single kernel generator $K \in E[d]$.

SQIsign uses x-only arithmetic on Montgomery curves throughout: points are represented by their x-coordinate (or projective $(X : Z)$ pairs), and the y-coordinate is avoided except where pairings require it. This follows the approach detailed in Costello and Hisil [CH17] and adopted in the SQIsign specification [SQI25a].

For an integer N coprime to p, the N-torsion subgroup $E[N] = \{P \in E(\overline{\mathbb{F}_p}) : [N]P = \mathcal{O}\}$ is isomorphic to $(\mathbb{Z}/N\mathbb{Z})^2$. A basis (P, Q) of $E[N]$ satisfies $E[N] = \langle P \rangle \oplus \langle Q \rangle$, and the Weil pairing $e_N(P, Q)$ is a primitive N-th root of unity. The SQIsign specification fixes a deterministic basis of $E[2^F]$ on each curve, generated with a one-byte hint by Algorithm 2.1 (TorsionBasisToHint) and rebuilt from the hint by Algorithm 2.2 (TorsionBasisFromHint). The choice of this basis is central to the compression formula of §3.

2.2 The SQIsign v2.0 signature scheme

We summarize the protocol at the level of detail needed to understand the compression; the full specification is in [SQI25a] and the underlying construction in [BDD+24]. A security proof appears in [ABDPW25].

SQIsign is a sigma protocol turned into a signature via the Fiat-Shamir transform [FS87]. The prover demonstrates knowledge of $\mathrm{End}(E_{\mathrm{pk}})$, which is computationally equivalent to knowing an isogeny path from the public base curve E_0 to the public key curve E_{pk}.

Key generation. The signer samples a random isogeny $\tau : E_0 \to E_{\mathrm{pk}}$ of smooth degree D_τ. The public key is the Montgomery coefficient A_{pk} of E_{pk} (64 bytes at Level I); the secret key is the ideal I_τ in $\mathrm{End}(E_0)$ corresponding to τ under the Deuring correspondence [DLLW23].

Signing. Given a message msg, the signer:

1. *Commits:* samples a random isogeny $\psi : E_0 \to E_{\mathrm{com}}$ and publishes E_{com}.
2. *Receives a challenge:* the verifier (or, via Fiat-Shamir, the hash function) produces a challenge isogeny $\varphi_{\mathrm{chl}} : E_{\mathrm{pk}} \to E_{\mathrm{chall}}$ of degree $2^{e_{\mathrm{chl}}}$ determined by the hash $\mathsf{chl} = \mathsf{HASH}(A_{\mathrm{pk}}, A_{\mathrm{com}}, \mathsf{msg})$.
3. *Responds:* using knowledge of $\mathrm{End}(E_{\mathrm{pk}})$, the signer finds an isogeny $\sigma : E_{\mathrm{com}} \to E_{\mathrm{chall}}$ of degree $q \approx \sqrt{p}$.

The critical change from v1.0 is step 3. The original SQIsign required σ to have smooth degree, found via the KLPT algorithm [DKLPW20, DLLW23]. SQIsign v2.0 [BDD+24] requires only that q be small (not necessarily smooth) and represents σ by its action on torsion points. A dimension-2 isogeny embedding via Kani's lemma [Kan97] makes that representation efficiently evaluable by the verifier.

Concretely, the signer constructs an auxiliary isogeny $\varphi_{\mathrm{aux}} : E_{\mathrm{com}} \to E_{\mathrm{aux}}$ such that $q + \deg(\varphi_{\mathrm{aux}}) = 2^{e_{\mathrm{rsp}}}$, and encodes σ as a 2×2 change-of-basis matrix $\mathbf{M}_{\mathrm{chl}}$ over $\mathbb{Z}/2^N\mathbb{Z}$. This matrix expresses, in the deterministic basis of E_{chall}, the images under the response of a basis of $E_{\mathrm{com}}[2^N]$. That basis of E_{com} is chosen so that the auxiliary isogeny maps it onto the deterministic basis of E_{aux} (Algorithm 4.8 of the specification; §3 makes this precise). The signature is

$$(\mathsf{chl},\ A_{\mathrm{aux}},\ \mathbf{M}_{\mathrm{chl}},\ n_{\mathrm{bt}},\ r_{\mathrm{rsp}},\ \mathsf{hint}_{\mathrm{aux}},\ \mathsf{hint}_{\mathrm{chall}}).$$

Verification. The verifier reconstructs E_{chall} from the public key and the challenge scalar, and regenerates the deterministic torsion bases on E_{chall} and E_{aux} from the transmitted hints. It applies $\mathbf{M}_{\mathrm{chl}}$ to obtain the torsion-point images of σ and constructs from them the Kani $(2^a, 2^a)$-isogeny $\Phi : E_{\mathrm{chall}} \times E_{\mathrm{aux}} \to E_{\mathrm{com}} \times E'_{\mathrm{aux}}$. It evaluates Φ to recover E_{com}, and checks that $\mathsf{chl} = \mathsf{HASH}(A_{\mathrm{pk}}, A_{\mathrm{com}}, \mathsf{msg})$.

Compression modifies only the first stage of this flow: recovering the full $\mathbf{M}_{\text{chl}}$ from a reduced representation before proceeding with the standard verification. Everything from the Kani embedding onward is unchanged.

2.3 Signature wire format

Table 3 of the specification [SQI25a] defines the standard signature as the tuple

$$(\mathsf{chl},\, A_{\text{aux}},\, \mathbf{M}_{\text{chl}},\, n_{\text{bt}},\, r_{\text{rsp}},\, \mathsf{hint}_{\text{aux}},\, \mathsf{hint}_{\text{chall}}).$$

At NIST Level I ($\lambda = 128$, $F = 248$, $e_{\text{rsp}} = 126$) the byte-level breakdown is as follows. A matrix entry occupies $\lceil (e_{\text{rsp}} + 2)/8 \rceil$ bytes, 16, 25 and 32 at the three levels, which is what the reference implementation and the sizes of Table 1 of the specification use.[1]

Field	Type	Bytes	Purpose
A_{aux}	$\mathbb{F}_{p^2}$ element	64	Montgomery coefficient of E_{aux}.
chl	integer	16	Challenge scalar; verifier recomputes φ_{chl} from this and E_{pk}.
M_{00}, M_{01}	scalars mod 2^N	2×16	First row of $\mathbf{M}_{\text{chl}}$: coefficients expressing $\sigma(P)$ in the canonical basis of E_{aux}.
M_{10}, M_{11}	scalars mod 2^N	2×16	Second row of $\mathbf{M}_{\text{chl}}$: coefficients for $\sigma(Q)$.
n_{bt}	integer	1	Number of backtracking steps in the response isogeny computation.
r_{rsp}	integer	1	$v_2(\deg \varphi_{\text{rsp}})$, the length of the even part of the response.
$\mathsf{hint}_{\text{aux}}$	byte	1	Disambiguates the deterministic basis of $E_{\text{aux}}[2^F]$.
$\mathsf{hint}_{\text{chall}}$	byte	1	Disambiguates the deterministic basis of $E_{\text{chl}}[2^F]$.
		148	

The matrix $\mathbf{M}_{\text{chl}}$ occupies 64 of the 148 bytes (43%). In FESTA [BMP23], QFESTA [NO24] and SQIsignHD [DLRW24] the determinant of the transmitted matrix is tied to the degree of the represented isogeny. The relation has the form $\det \mathbf{M} \equiv k \cdot q$, with k fixed by the pairings of the bases. In SQIsign v2.0 the auxiliary isogeny takes the place of q and the relation becomes the one of Theorem 3.2. In the standard format the specification transmits all four entries and uses no such relation.

At Levels III and V the field sizes scale with λ, producing 224-byte and 292-byte signatures respectively. The matrix fraction remains roughly 43% at all levels.

2.4 The Weil pairing

For background on pairings over elliptic curves we refer to Silverman [Sil86]; for the specific computational methods used in SQIsign we refer to Robert [Rob24] and Pope et al. [PRRS+25].

[1] Appendix B of [SQI25a] prints the entry size as $\lfloor (e_{\text{rsp}} + 7)/8 \rfloor$, which gives 24 at level III and a 220-byte signature, against the 224 of its Table 1. The implementation's $\lceil (e_{\text{rsp}} + 2)/8 \rceil = 25$ is the one consistent with the table.

Let N be a positive integer coprime to $\mathrm{char}(\mathbb{F}_q)$. The Weil pairing $e_N : E[N] \times E[N] \to \mu_N$ maps pairs of N-torsion points to N-th roots of unity in $\mathbb{F}_{q^k}^*$, where k is the embedding degree. Three of its properties matter for the recovery formula:

(i) *Bilinearity:* $e_N(P_1+P_2, Q) = e_N(P_1, Q)\, e_N(P_2, Q)$, and symmetrically in the second argument.
(ii) *Antisymmetry:* $e_N(P, Q) = e_N(Q, P)^{-1}$.
(iii) *Non-degeneracy:* if (P, Q) is a basis of $E[N]$, then $e_N(P, Q)$ is a primitive N-th root of unity.

Property (ii) distinguishes the Weil pairing from the Tate pairing, which satisfies $\hat{t}_N(P, Q) = \hat{t}_N(Q, P)$ (symmetry, not antisymmetry) after the final exponentiation. The relationship between the two pairings, and the role of the torsion order in determining whether the Tate pairing can substitute for the Weil pairing in determinant recovery, is analyzed in §6.

The SQIsign specification computes pairings via a biextension cubical ladder on Montgomery curves (Section 8.3 of [SQI25a]). Pope et al. [PRRS+25], building on Robert [Rob24], showed that this approach yields a >40% speedup over Miller's algorithm for the pairing orders used in SQIsign. The specification uses the Tate pairing for change-of-basis computations during signing (Section 2.2.5.2 of [SQI25a]). It notes that "one may also use the Weil pairing to compute the change of basis, although at an increase in cost."

3 The decompression formula

3.1 What the verifier holds

Fix a public key $(E_{\mathrm{pk}}, \mathsf{hint}_{\mathrm{pk}})$ and a signature $\sigma = (E_{\mathrm{aux}}, n_{\mathrm{bt}}, r_{\mathrm{rsp}}, \mathbf{M}, \mathsf{chl}, \mathsf{hint}_{\mathrm{aux}}, \mathsf{hint}_{\mathrm{chl}})$ accepted by Algorithm 4.9 of the specification [SQI25a]. Write $e' = e_{\mathrm{rsp}} - r_{\mathrm{rsp}} - n_{\mathrm{bt}}$, $N = e' + r_{\mathrm{rsp}} + 2$ and $g = N - 2$ as in §1.2. The verifier computes the following objects.

(i) E_{chl} from E_{pk} and chl (line 9), the deterministic basis $(P_{\mathrm{chl}}^{\mathrm{det}}, Q_{\mathrm{chl}}^{\mathrm{det}})$ of $E_{\mathrm{chl}}[2^F]$ from $\mathsf{hint}_{\mathrm{chl}}$ (line 12), and $A = [2^{F-N}]P_{\mathrm{chl}}^{\mathrm{det}}$, $B = [2^{F-N}]Q_{\mathrm{chl}}^{\mathrm{det}}$ (line 13), a basis of $E_{\mathrm{chl}}[2^N]$.
(ii) The deterministic basis of $E_{\mathrm{aux}}[2^F]$ from $\mathsf{hint}_{\mathrm{aux}}$ (line 10), and $C = [2^{F-e'-2}]P_{\mathrm{aux}}^{\mathrm{det}}$, $D = [2^{F-e'-2}]Q_{\mathrm{aux}}^{\mathrm{det}}$ (line 11), a basis of $E_{\mathrm{aux}}[2^{e'+2}]$.
(iii) $P = [M_{00}]A + [M_{10}]B$ and $Q = [M_{01}]A + [M_{11}]B$ (line 14).

The entries are transmitted in the order $M_{00}, M_{01}, M_{10}, M_{11}$ (Section 4.6 of the specification), and (iii) is how the reference implementation applies them: the first column acts on the first basis point. Section 1.4 of the specification writes the same product with (A, B) as a column vector, which transposes the index names. Nothing below depends on the choice except the names of the pivot entries in §4, which follow the implementation.

Define

$$\omega = e_{2^N}(A, B) \in \mu_{2^N}, \qquad \eta = e_{2^{e'+2}}(C, D) \in \mu_{2^{e'+2}} \subseteq \mu_{2^N}.$$

Both are primitive roots of unity of their orders, since each is the Weil pairing of a basis, and $\mu_{2^N} = \langle \omega \rangle$.

3.2 What the matrix records

The signer (Algorithm 4.2 of the specification) produces a response isogeny $\varphi_{\mathrm{rsp}} = \varphi_{\mathrm{rsp}}^{\mathrm{even}} \circ \varphi_{\mathrm{rsp}}^{\mathrm{odd}} : E_{\mathrm{com}} \to E_{\mathrm{chl}}$ with $\deg \varphi_{\mathrm{rsp}}^{\mathrm{odd}} = q$ odd (line 18) and $\deg \varphi_{\mathrm{rsp}}^{\mathrm{even}} = 2^{r_{\mathrm{rsp}}}$ (line 17 and Algorithm 4.6), and an auxiliary isogeny $\varphi_{\mathrm{aux}} : E_{\mathrm{com}} \to E_{\mathrm{aux}}$ of degree $2^{e'} - q$ (line 23). Let $R_0 = [2^{F-N}]\varphi_{\mathrm{com}}(P_0)$ and $S_0 = [2^{F-N}]\varphi_{\mathrm{com}}(Q_0)$; since $\deg \varphi_{\mathrm{com}}$ is odd, (R_0, S_0) is a basis of $E_{\mathrm{com}}[2^N]$. The points the signer carries into Algorithm 4.8 are (page 42 of the specification, after Algorithms 4.6 and 4.7)

$$(P_{\mathrm{chl}}, Q_{\mathrm{chl}}) = (\varphi_{\mathrm{rsp}}(R_0), \varphi_{\mathrm{rsp}}(S_0)), \qquad (P_{\mathrm{aux}}, Q_{\mathrm{aux}}) = (\varphi_{\mathrm{aux}}(R_0), \varphi_{\mathrm{aux}}(S_0)),$$

all of order 2^N. Algorithm 4.8 computes a matrix $\mathbf{M}_1$ that expresses the deterministic basis of $E_{\mathrm{aux}}[2^N]$ in $(P_{\mathrm{aux}}, Q_{\mathrm{aux}})$ (line 5), applies $\mathbf{M}_1$ to $(P_{\mathrm{chl}}, Q_{\mathrm{chl}})$ (line 6), and expresses the result in the deterministic basis of $E_{\mathrm{chl}}[2^N]$. That matrix is $\mathbf{M}_{\mathrm{chl}}$ (line 7). Put $(R, S) = \mathbf{M}_1 \cdot (R_0, S_0)$, again a basis of $E_{\mathrm{com}}[2^N]$ because $\mathbf{M}_1$ is invertible. Isogenies are linear, so:

Lemma 3.1. *For a signature produced by Algorithm 4.2 there is a basis (R, S) of $E_{\mathrm{com}}[2^N]$ such that*

$$P = \varphi_{\mathrm{rsp}}(R), \quad Q = \varphi_{\mathrm{rsp}}(S), \quad C = \varphi_{\mathrm{aux}}([2^{r_{\mathrm{rsp}}}]R), \quad D = \varphi_{\mathrm{aux}}([2^{r_{\mathrm{rsp}}}]S).$$

Proof. With (R, S) as above, the deterministic basis of $E_{\mathrm{aux}}[2^N]$ is $(\varphi_{\mathrm{aux}}(R), \varphi_{\mathrm{aux}}(S))$ by line 5 of Algorithm 4.8. Since $2^{F-e'-2} = 2^{r_{\mathrm{rsp}}} \cdot 2^{F-N}$, the pair (C, D) is its $[2^{r_{\mathrm{rsp}}}]$-multiple by (ii) of §3.1. Lines 6 and 7 of Algorithm 4.8 say that $\mathbf{M}_{\mathrm{chl}}$ applied to the deterministic basis of $E_{\mathrm{chl}}[2^N]$ is $(\varphi_{\mathrm{rsp}}(R), \varphi_{\mathrm{rsp}}(S))$, which is (iii). □

3.3 Pairing facts

We use the following properties of the Weil pairing; see Silverman [Sil86, III.8.1]. For n coprime to p, $T, U \in E[n]$ and an isogeny φ:

(W1) bilinearity;
(W2) alternation, $e_n(T, T) = 1$, hence $e_n(T, U) = e_n(U, T)^{-1}$;
(W3) compatibility with isogenies, $e_n(\varphi(T), \varphi(U)) = e_n(T, U)^{\deg \varphi}$;
(W4) change of level: for $T \in E[mn]$ and $U \in E[n]$, $e_{mn}(T, U) = e_n([m]T, U)$;
(W5) non-degeneracy: if (T, U) is a basis of $E[n]$ then $e_n(T, U)$ has order exactly n.

3.4 Main result

Theorem 3.2 (Weil pairing determinant recovery)**.** *For a signature produced by Algorithm 4.2 of [SQI25a], with ω, η and $\mathbf{M}$ as in §3.1,*

$$\omega^{\det \mathbf{M}} \cdot \eta \;=\; \zeta \qquad \text{with } \zeta \text{ a primitive fourth root of unity.}$$

Consequently, with $\log_\omega$ the discrete logarithm in $\mu_{2^N} = \langle \omega \rangle$,

$$\det(\mathbf{M}) \;\equiv\; \log_\omega\!\left(\eta^{-1}\right) \pmod{2^g}, \qquad g = e_{\mathrm{rsp}} - n_{\mathrm{bt}}.$$

The modulus is 2^g, the base is $\omega = e_{2^N}(A, B)$, and the argument is η^{-1}, not η. The entries are indexed as the reference implementation applies them, $P = [M_{00}]A + [M_{10}]B$; Section 1.4 of the specification writes the product with (A, B) as a column vector, which transposes the index names and changes nothing else.

Proof. Step 1 (the matrix). By (W1) and (W2),

$$\begin{aligned} e_{2^N}(P, Q) &= e_{2^N}([M_{00}]A + [M_{10}]B,\ [M_{01}]A + [M_{11}]B) \\ &= e_{2^N}(A, B)^{M_{00}M_{11}}\, e_{2^N}(B, A)^{M_{10}M_{01}} = \omega^{\det \mathbf{M}}. \end{aligned}$$

Step 2 (the response). Let $\rho = e_{2^N}(R, S)$ for the basis of Lemma 3.1; by (W5) it has order exactly 2^N. By the lemma and (W3),

$$e_{2^N}(P, Q) = e_{2^N}(\varphi_{\mathrm{rsp}}(R), \varphi_{\mathrm{rsp}}(S)) = \rho^{\deg \varphi_{\mathrm{rsp}}} = \rho^{2^{r_{\mathrm{rsp}}} q}.$$

Step 3 (the auxiliary isogeny). By the lemma and (W3), $\eta = e_{2^{e'+2}}([2^{r_{\mathrm{rsp}}}]R, [2^{r_{\mathrm{rsp}}}]S)^{2^{e'}-q}$. By (W4) with $m = 2^{r_{\mathrm{rsp}}}$ and $n = 2^{e'+2}$, and then (W1),

$$e_{2^{e'+2}}([2^{r_{\mathrm{rsp}}}]R, [2^{r_{\mathrm{rsp}}}]S) = e_{2^N}(R, [2^{r_{\mathrm{rsp}}}]S) = \rho^{2^{r_{\mathrm{rsp}}}}, \qquad \text{so} \qquad \eta = \rho^{2^{r_{\mathrm{rsp}}}(2^{e'}-q)}.$$

Step 4 (the Kani square). Multiplying Steps 1 to 3,

$$\omega^{\det \mathbf{M}} \cdot \eta = \rho^{2^{r_{\mathrm{rsp}}} q + 2^{r_{\mathrm{rsp}}}(2^{e'}-q)} = \rho^{2^{e'+r_{\mathrm{rsp}}}} = \rho^{2^{N-2}} =: \zeta.$$

Since ρ has order 2^N, ζ has order exactly 4.

Step 5 (the logarithm). Since $\eta^{-1} \in \mu_{2^N} = \langle \omega \rangle$, the integer $y = \log_\omega(\eta^{-1}) \in \mathbb{Z}/2^N\mathbb{Z}$ is defined, and $\zeta = \omega^{2^{N-2}u}$ for some $u \in \{1, 3\}$. Step 4 reads $\omega^{\det \mathbf{M}} = \omega^y \, \omega^{2^{N-2}u}$, that is

$$\det \mathbf{M} \;\equiv\; y + 2^{N-2}u \pmod{2^N}, \qquad u \in \{1, 3\}. \tag{1}$$

Reducing modulo $2^{N-2} = 2^g$ gives the formula. □

Remark 3.3 (Where the degree q goes)**.** *Steps 2 and 3 carry q and $2^{e'} - q$. Their sum $2^{e'}$ is the condition the signer imposes when it samples the auxiliary ideal of norm $2^{e'} - q$ (Algorithm 4.2, line 23, and Figure 4 of the specification). That condition is what removes q from the formula. In SQIsignHD the auxiliary side carries no isogeny and the corresponding relation reads $a_1 b_2 - b_1 a_2 \equiv kq$ [DLRW24, Section 6.1]. In SQIsign v2.0 the auxiliary isogeny absorbs the degree.*

Remark 3.4 (Why η appears inverted)**.** *The relation of Step 4 is a product, so η moves to the other side as η^{-1} while the base of the logarithm is ω. The discrete logarithm routine of the reference implementation takes the inverse of its base as an argument; the base is ω.*

Remark 3.5 (Isotropy alone is weaker)**.** *Line 24 of Algorithm 4.9 checks that the pair $([4]P, [4]C)$, $([4]Q, [4]D)$ is isotropic, which by (W1) to (W4) is the identity $(\omega^{\det \mathbf{M}}\eta)^{16} = 1$. A forged kernel that passes this check pins* $\det \mathbf{M}$ *modulo 2^{N-4} only. The modulus 2^{N-2} of the theorem comes from Step 4, the Kani square of an honest signer, not from isotropy.*

Corollary 3.6 (Valuation of the determinant)**.** *For a signature produced by Algorithm 4.2, $v_2(\det \mathbf{M}) = r_{\mathrm{rsp}}$. In particular* $\det \mathbf{M}$ *is odd if and only if $r_{\mathrm{rsp}} = 0$.*

Proof. ω and ρ both generate μ_{2^N}, so $\rho = \omega^t$ with t odd. Steps 1 and 2 give $\det \mathbf{M} \equiv t \cdot 2^{r_{\mathrm{rsp}}} q \pmod{2^N}$ with tq odd and $r_{\mathrm{rsp}} < N$. □

3.5 Precision

Proposition 3.7. *Theorem 3.2 determines* $\det \mathbf{M}$ *modulo 2^g, while the specification carries it modulo 2^N. The difference $N - g$ equals 2 identically. If the logarithm y of Step 5 is taken in μ_{2^N}, equation* (1) *leaves exactly one bit of* $\det \mathbf{M}$ *undetermined, bit $N - 1$; bit $N - 2$ of* $\det \mathbf{M} - y$ *is always* 1*.*

Proof. $N = e' + r_{\mathrm{rsp}} + 2$ is the bound on the entries in line 5 of Algorithm 4.9, and $g = e' + r_{\mathrm{rsp}} = e_{\mathrm{rsp}} - n_{\mathrm{bt}}$ by the definition of e' (Algorithm 4.2, line 20), so $N - g = 2$. The 2 is the extra torsion of the $(2, 2)$-isogeny chain: the specification feeds Isogeny22ChainWithTorsion isotropic points of order $2^{e'+2}$ for a chain of length e' (Section 2.4.1), and the reference implementation names the constant `HD_EXTRA_TORSION`. The statement about the bits is equation (1) with u odd. □

The compressed format of §5 transmits one bit of the dropped matrix entry, its bit $N-1$. Solving $M_{00}x \equiv \det \mathbf{M} + M_{01}M_{10} \pmod{2^N}$ for the dropped entry x with $\det \mathbf{M}$ from (1) gives two candidates that differ by 2^{N-1} times an odd number. So bits 0 to $N-2$ of x are determined and only bit $N-1$ is free. On the 300 KAT vectors the quotient u is 1 on 152 and 3 on 148 and never even (§3.6).

3.6 Numerical confirmation

We checked every step on the 300 official round-2 KAT vectors of the reference [SQI25c] at tag `nist-v2`, 100 per level, with the verifier of [Ray26b] at tag `v0.4.31`.

- The bases: the decompressor's (A, B) has order 2^N and (C, D) order $2^{e'+2}$; ω and η have exactly those orders on 100/100 vectors per level.
- Step 1: $e_{2^N}(P, Q) = \omega^{\det \mathbf{M}}$ on 100/100 per level, with P and Q built by the verifier's own scalar multiplication.
- Step 4: $\omega^{\det \mathbf{M}}\eta$ equals 1 on 0, -1 on 0, and has square -1 on 100 of 100 vectors per level: a primitive fourth root of unity on all 300. The same holds after the even part is pushed through, at order $2^{e'+2}$, on every vector with $r_{\mathrm{rsp}} > 0$ (52, 61 and 53 of 100).
- The formula: with $y = \log_\omega(\eta^{-1})$ computed by the implementation's discrete logarithm at order 2^N, the lowest bit in which y and $\det \mathbf{M}$ differ is bit $g = e_{\mathrm{rsp}} - n_{\mathrm{bt}}$ on 100/100 vectors per level, never lower and never higher; it depends on n_{bt} and not on r_{rsp}. The three other sign variants ($\log_\omega(\eta)$ and the two with base ω^{-1}) agree with $\det \mathbf{M}$ on 0 of 100. No correction by the cofactor c or by a power of 2 up to 16 holds on any vector.
- Equation (1): $(\det \mathbf{M} - y) \bmod 2^N$ is 2^{N-2} on 48, 49 and 55 vectors and $3 \cdot 2^{N-2}$ on 52, 51 and 45, at levels I, III and V; never 0 and never $2 \cdot 2^{N-2}$.
- Corollary 3.6: $v_2(\det \mathbf{M}) = r_{\mathrm{rsp}}$ on 100/100 per level, over r_{rsp} from 0 to 7, 8 and 5 at the three levels.

3.7 Pairing selection

The formula needs a pairing that is bilinear and alternating on the bases it is evaluated on, so that Step 1 of the proof reads off the determinant. The Weil pairing has both properties at every order. The reduced Tate pairing is bilinear, and on the maximal supersingular curves of SQIsign it is alternating as well whenever its order divides $p+1$ (Section 2.2.5.1 of the specification). What it lacks on the scaled bases (A, B) is order: §6.1 derives and measures how many bits its values carry.

4 Adaptive pivot and matrix recovery

Given $\det(\mathbf{M}) \bmod 2^{N-1}$ from Theorem 3.2 and Proposition 3.7 and three of the four matrix entries, the fourth is recovered modulo 2^{N-1} by solving $M_{00}M_{11} - M_{01}M_{10} \equiv \det(\mathbf{M}) \pmod{2^{N-1}}$. Its top bit comes from the hint of §4.3. The choice of which entry to drop depends on invertibility modulo 2^{N-1}. Dartois et al. [DLRW24] note the analogous condition for SQIsignHD: "If a_1 is invertible, we can recover b_2 from the other scalars ... and we can recover a_2 otherwise." We make this concrete for the v2.0 wire format, quantify the frequency of each case, and describe the detection mechanism. The entry names follow §3.1: M_{00} and M_{01} are the coefficients of A in P and in Q.

4.1 Standard recovery when M_{00} is odd

The compressed signature transmits M_{00}, M_{01}, and one of $\{M_{10}, M_{11}\}$. In the common case the compressor drops M_{11} and the decompressor recovers it as

$$M_{11} \;\equiv\; \big(\det(\mathbf{M}) + M_{01} \cdot M_{10}\big) \cdot M_{00}^{-1} \pmod{2^{N-1}}, \tag{2}$$

where the modular inverse $M_{00}^{-1} \bmod 2^{N-1}$ exists whenever M_{00} is odd. The inverse is computed by Newton iteration: starting from $M_{00}^{-1} \bmod 2$ (which is 1, since M_{00} is odd) and iterating $x \leftarrow x(2 - M_{00}\,x) \bmod 2^{2^i}$ for $i = 1, \ldots, \lceil \log_2(N-1) \rceil$. This costs $O(\log N)$ multiplications modulo 2^{N-1}.

4.2 Adaptive pivot when M_{00} is even

When M_{00} is even it is a zero divisor modulo 2^{N-1} and Equation (2) has no solution.

Proposition 4.1. *For a signature with $r_{\mathrm{rsp}} = 0$, $\det(\mathbf{M})$ is odd, and if M_{00} is even then M_{01} is odd.*

Proof. Corollary 3.6 gives $v_2(\det \mathbf{M}) = r_{\mathrm{rsp}} = 0$. If M_{00} and M_{01} were both even, then $\det \mathbf{M} = M_{00}M_{11} - M_{01}M_{10}$ would be even. □

When $r_{\mathrm{rsp}} > 0$ the determinant is even (Corollary 3.6) and the proposition does not apply. The pivot still exists, for a different reason.

Proposition 4.2. *For a signature produced by Algorithm 4.2, M_{00} and M_{01} are not both even.*

Proof. For $r_{\mathrm{rsp}} = 0$ this is Proposition 4.1. Let $r_{\mathrm{rsp}} \geq 1$ and write $\Lambda = \langle P, Q \rangle = \varphi_{\mathrm{rsp}}(E_{\mathrm{com}}[2^N])$. Both M_{00} and M_{01} even means $\Lambda \subseteq \langle [2]A, B \rangle = \langle B \rangle + [2]E_{\mathrm{chl}}[2^N]$. Multiplication by 2^{N-1} maps $E_{\mathrm{chl}}[2^N]/[2]E_{\mathrm{chl}}[2^N]$ isomorphically onto $E_{\mathrm{chl}}[2]$, so the condition says $[2^{N-1}]\Lambda \subseteq \langle [2^{N-1}]B \rangle$. Now $[2^{N-1}]\Lambda = \varphi_{\mathrm{rsp}}^{\mathrm{even}}(E[2])$ is the group generated by the one 2-torsion point T in the kernel of the dual of $\varphi_{\mathrm{rsp}}^{\mathrm{even}}$, and $[2^{N-1}]B = (0,0)$ because the deterministic basis has its second point above $(0,0)$ (Section 2.2.3 of the specification). So the condition says $T = (0,0)$. But $(0,0)$ on E_{chl} generates the kernel of the dual of the last step of the challenge chain (Section 2.3.1, applied by Algorithm 4.7). If $T = (0,0)$, the response's even part and the challenge end with the same 2-isogeny, so the endomorphism of E_0 that the signer normalizes in Algorithm 4.4 factors through [2], which that normalization excludes. □

On the 300 KAT vectors M_{00} and M_{01} are both even on 0 of 100 per level, while M_{00} and M_{10} are both even on 14, 21 and 18 of 100. Those are exactly the vectors where the verifier takes its even-part kernel from Q rather than P (line 16 of Algorithm 4.9, in the implementation's column convention).

When M_{00} is even the compressor drops M_{10} instead of M_{11}, and the decompressor recovers it as

$$M_{10} \;\equiv\; \big(M_{00} \cdot M_{11} - \det(\mathbf{M})\big) \cdot M_{01}^{-1} \pmod{2^{N-1}}. \tag{3}$$

The decompressor selects between Equations (2) and (3) by testing the parity of the transmitted M_{00}: odd means M_{11} was dropped, even means M_{10} was. This requires no additional wire-format bits; the pivot path is determined by data already in the compressed signature.

On the official KAT vectors M_{00} is even on 28, 37 and 36 of 100 at levels I, III and V, so the pivot path is taken on about a third of the signatures. The entries M_{01}, M_{10} and M_{11} are even on 33, 42, 56 (level I), 33, 57, 52 (level III) and 37, 53, 54 (level V) of 100. The valuation of the determinant, r_{rsp}, is 0 on 48, 39 and 47 of 100 and at most 8.

4.3 The one-bit hint

The recovery of Equations (2) and (3) determines the dropped entry modulo 2^{N-1}, and the entry has N bits. The compressor therefore transmits the top bit of the dropped entry x,

$$h \;=\; \lfloor x/2^{N-1} \rfloor,$$

in the metadata byte (§5.3). By Proposition 3.7 it carries the same information as the top bit of u in (1); encoding u itself would need the two pairings at compression time, and the entry bit does not. Verification itself never reads bit $N-1$ of an entry, since the chain of Algorithm 4.9 uses the points only through their multiples (§8 gives the argument); the hint buys a byte-identical round trip, not acceptance.

Over the 100 level I KAT vectors h is 0 on 52 and 1 on 48; §6.2 gives the counts at every level and against n_{bt}, r_{rsp} and the parity of M_{00}.

The hint cannot be removed by signer retry at acceptable cost. Each signing attempt produces a target value of h with probability $1/2$ if h is uniform. A fixed value with 95% confidence then needs 5 attempts, since $2^{-5} < 0.05$: a fivefold signing slowdown for one bit. No relation we know fixes it: it is the top bit of u, a property of the signer's basis of E_{com} that the verifier never sees.

5 Compression from 148 to 129 bytes

This section assembles the techniques from §3 and §4 into a concrete compressed wire format. Each step is independent: an implementation may adopt any prefix of the sequence below without committing to the later steps.

5.1 Overview

Size (L1)	Technique	Section
148 B	Standard format (baseline)	§2.3
133 B	Drop one matrix entry + Weil pairing recovery	§3
132 B	Precision analysis: a one-bit hint, not $2 + r_{\mathrm{rsp}}$ bits	§3
130 B	Recompute canonical basis hints from curves	§5.2
129 B	Pack metadata into 1 byte	§5.3

The first step accounts for 15 of the 19 saved bytes: dropping one 16-byte matrix entry and adding a 1-byte hint gives a net saving of 15 bytes. The precision analysis shrinks the hint from a full byte to one bit, saving 1 byte. Hint recomputation and metadata packing each save 1 byte.

5.2 Recomputing hints from curves

The standard format includes two 1-byte hints, $\mathsf{hint}_{\mathrm{aux}}$ and $\mathsf{hint}_{\mathrm{chall}}$, which disambiguate the deterministic basis of $E[2^F]$ on E_{aux} and E_{chall} respectively. These hints are outputs of Algorithm 2.1 (TorsionBasisToHint) in the specification [SQI25a] and depend only on the Montgomery coefficient of the curve.

During decompression both curves are already available: E_{chall} is reconstructed from the public key and the challenge scalar, and E_{aux} is given by the transmitted A_{aux}. The decompressor can therefore call TorsionBasisToHint on each curve and recover both hints deterministically. This is the same computation the verifier would perform regardless; recomputing the hints merely moves it

ahead of matrix recovery rather than after. The two hint bytes are thus redundant in the presence of the curve data and can be dropped, saving 2 bytes.

Verified on all 300 KAT vectors: the recomputed hints match the transmitted hints on 100 of 100 at each level, for both curves.

5.3 Metadata packing

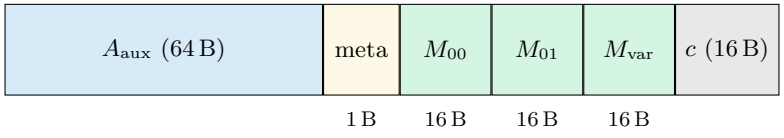
The standard format allocates 1 byte each to n_{bt} and r_{rsp}, even though their ranges are small: on the 300 KAT vectors n_{bt} is at most 4 and r_{rsp} at most 8 (§6.2). The compressed format packs them with the one-bit hint of §4.3 into a single metadata byte that allows $n_{\mathrm{bt}} \leq 7$ and $r_{\mathrm{rsp}} \leq 15$. The compressor rejects larger values rather than truncating them:

Bits 7 to 4	Bit 3	Bits 2 to 0	Total
r_{rsp} (4 bits)	h (1 bit)	n_{bt} (3 bits)	8 bits

This layout puts r_{rsp} in the high nibble, read by a right shift, the hint in bit 3, and n_{bt} in the low 3 bits, read by masking with `0x07`. The packing replaces 2 bytes of metadata with 1 byte, saving 1 byte.

5.4 Wire format

The compressed signature at Level I is 129 bytes with the following layout. The first row of $\mathbf{M}_{\mathrm{chl}}$ (M_{00} and M_{01}) is always transmitted. The third matrix field M_{var} contains M_{10} if M_{00} is odd (standard path) or M_{11} if M_{00} is even (pivot path); the decompressor infers the path from the parity of the transmitted M_{00}.



Fields not labeled are fixed-size: A_{aux} is always 64 bytes and chl is always 16 bytes at Level I. The metadata byte encodes r_{rsp}, h, and n_{bt} as described in §5.3. No challenge hints, no auxiliary hints, no separate n_{bt} or r_{rsp} bytes.

5.5 Sizes across security levels

Format	Level I	Level III	Level V
Standard	148 B	224 B	292 B
Compressed	129 B	196 B	257 B
Savings	19 B (13%)	28 B (12%)	35 B (12%)

The savings grow in bytes at higher levels because the dropped entry grows with λ. They shrink as a fraction because the A_{aux} field, which these techniques do not touch, is a larger part of the total.

5.6 Security considerations

The compressed format is a deterministic, invertible transformation of the standard format. Given a standard signature, compression produces a unique compressed signature, and decompression recovers the original matrix $\mathbf{M}_{\text{chl}}$ exactly. On the KAT vectors the decompressed signature is byte-identical to the original on 100 of 100 at each level. The verifier detects a failed decompression because the recovered matrix must produce a valid Kani $(2,2)$-isogeny. An incorrect $\det(\mathbf{M})$ or a corrupted matrix entry makes the isogeny construction fail or the final hash check reject. Aardal et al. [ABDPW25] prove the security of SQIsign v2.0 for the standard signature format. Decompression is a preprocessing step that either reproduces the standard-format matrix exactly or causes verification to reject, so the security reduction applies to the compressed format without modification.

6 Pairing selection and negative results

We analyze the choice of pairing (§6.1), report the distributions behind the metadata byte (§6.2), and document two approaches to further compression that turn out to need changes to the signer (§6.3).

6.1 Tate pairing: degeneracy and the order of its values

The Tate pairing costs one cubical ladder where the Weil pairing costs two, and the specification uses it for its own change-of-basis computation (Algorithm 2.5). This section derives what one Tate ladder per curve can recover, and measures it.

The transmitted basis is divisible. The verifier does not hold a basis of $E_{\text{chl}}[2^N]$ chosen at random; it holds $A = [2^{F-N}]P^{\text{det}}$ and $B = [2^{F-N}]Q^{\text{det}}$ for points of order 2^F (§3.1). The reduced Tate pairing of order n is a map $t_n : E[n] \times E(\mathbb{F}_{p^2})/nE(\mathbb{F}_{p^2}) \to \mu_n$ (Section 2.2.5.1 of the specification). For the curves at hand $E(\mathbb{F}_{p^2}) \cong (\mathbb{Z}/(p+1)\mathbb{Z})^2$, whose 2-primary part is $(\mathbb{Z}/2^F\mathbb{Z})^2$, so the 2-primary part of the quotient by $2^N E(\mathbb{F}_{p^2})$ is $(\mathbb{Z}/2^N\mathbb{Z})^2$. In it the class of $B = [2^{F-N}]Q^{\text{det}}$ has order $2^{N-(F-N)} = 2^{2N-F}$ when $2N > F$, and is trivial otherwise. Hence

$$t_{2^N}(A, B) \in \mu_{2^{2N-F}}, \qquad \text{and likewise } t_{2^N}(B, A).$$

At level I, $F = 248$ and $N = 128 - n_{\text{bt}}$, so the value has order at most 2^8 against the $g = 126 - n_{\text{bt}}$ bits the recovery needs. The final exponentiation $(p^2-1)/2^N$ has 2-adic valuation $F - N + 1$; the loss is in the argument, not in the exponent.

Lengthening the ladder does not help. For T of order 2^N and $m \geq N$, the function $f_{2^m,T}$ with divisor $2^m(T) - 2^m(0_E)$ is $f_{2^N,T}^{2^{m-N}}$, so $t_{2^m}(T, U) = f_{2^N,T}(U)^{2^{m-N}(p^2-1)/2^m} = t_{2^N}(T, U)$. A ladder of 2^m steps on the scaled points computes the same value for every $m \geq N$. On the KAT vectors the values, their orders and the bits they recover are identical for every m from N to F.

Full-order points. The verifier also holds the points of order 2^F from which (A, B) is scaled, and can scale them to any order 2^m with $N \leq m \leq F$: $A_m = [2^{F-m}]P^{\text{det}}$, $B_m = [2^{F-m}]Q^{\text{det}}$, and likewise (C_m, D_m) on E_{aux}. The same argument gives $t_{2^m}(A_m, B_m) \in \mu_{2^{2m-F}}$. Measured on the 300 KAT vectors for every m from N to F, in both argument orders and on both curves, the order is exactly $2^{\max(2m-F,0)}$.

The reduced Tate pairing is alternating here. On a maximal supersingular curve the reduced Tate pairing of order n is alternating whenever n divides $p+1$. Section 2.2.5.1 of the specification states: "When $E/\mathbb{F}_{p^2}$ is a maximal supersingular curve, and $n \mid p+1$, then the reduced Tate-Lichtenbaum pairing is alternate." So $t_n(T,T) = 1$ and $t_n(U,T) = t_n(T,U)^{-1}$, and Step 1 of the proof of Theorem 3.2 reads off $\det \mathbf{M}$ from it exactly as from the Weil pairing. Moreover the Weil pairing is the ratio $t_n(T,U)/t_n(U,T)$ of the two Miller values before the final exponentiation (Section 8.3), so after it

$$t_n(T,U)^2 \;=\; e_n(T,U)^{(p^2-1)/n}, \qquad \frac{p^2-1}{2^m} = 2^{F-m+1}\cdot w, \quad w = c\,(c\,2^{F-1}-1) \text{ odd.} \tag{4}$$

This is where the cofactor c enters, and only there: it also multiplies the hint points when the deterministic basis is built (Algorithm 2.1, lines 13 and 14), but both uses leave the odd exponent w, which is a unit modulo 2^g and common to the two curves.

Proposition 6.1 (recovery from one Tate ladder per curve)**.** *Let $N \le m \le F$, let $\tau = t_{2^m}(A_m, B_m)$ and $\tau' = t_{2^m}(C_m, D_m)$, and let $y_m = \log_\tau(\tau'^{-1})$ in the group generated by τ, which has order 2^{2m-F}. Then*

$$\det \mathbf{M} \;\equiv\; 2^{r_{\mathrm{rsp}}}\, y_m \pmod{2^{\min(g,\, 2m-F-1+r_{\mathrm{rsp}})}}.$$

Proof. Let $\varepsilon = e_{2^F}(R,S)$ and $\varepsilon' = e_{2^F}(R',S')$ for the full-order deterministic bases of the two curves, both of order 2^F, and $\lambda = \log_\varepsilon(\varepsilon'^{-1})$. By (W4) and (W1), $\omega = \varepsilon^{2^{F-N}}$, $\eta = \varepsilon'^{2^{F-N+r_{\mathrm{rsp}}}}$ and $e_{2^m}(A_m,B_m) = \varepsilon^{2^{F-m}}$, so by (4) $\tau^2 = \varepsilon^{2^{2F-2m+1}w}$ and $\tau'^2 = \varepsilon'^{2^{2F-2m+1}w}$. Squaring $\tau'^{-1} = \tau^{y_m}$ and comparing exponents of ε gives $\lambda \equiv y_m \pmod{2^{2m-F-1}}$. Step 4 of Theorem 3.2 in terms of ε reads $\varepsilon^{2^{F-N}(\det\mathbf{M} - 2^{r_{\mathrm{rsp}}}\lambda)} = \zeta$ with ζ of order 4, hence $\det\mathbf{M} \equiv 2^{r_{\mathrm{rsp}}}\lambda \pmod{2^g}$. Combining the two congruences gives the claim. □

The bit lost in the squaring is the sign of the square root in (4). On the KAT vectors it is not lost: the congruence holds modulo $2^{\min(g,2m-F+r_{\mathrm{rsp}})}$ on 300 of 300, so the two curves' Tate values take the same sign. One Tate ladder per curve therefore recovers $\det\mathbf{M}$ as soon as $2m-F \ge g-r_{\mathrm{rsp}}$, that is for

$$m \;\ge\; \left\lceil \tfrac{F+g-r_{\mathrm{rsp}}}{2} \right\rceil,$$

which for $n_{\mathrm{bt}} = r_{\mathrm{rsp}} = 0$ is 187, 284 and 377 at the three levels.

Measurement. Table 1 summarizes the sweep over every m from N to F on the 300 KAT vectors. The per-vector rule $m \ge \lceil (F+g-r_{\mathrm{rsp}})/2 \rceil$ holds with equality on 300 of 300, and the comparison with $-\det\mathbf{M}$ fails on every vector at every m.

Table 1: One Tate ladder per curve on the deterministic bases scaled to order 2^m: vectors (of 100 per level) on which the logarithm recovers $\det\mathbf{M}$ modulo 2^g, by m. On the transmitted bases of order 2^N the count is 0 at every m.

Level	**order of the value**	**vectors recovered, by m**	**all 100 from**
I ($F=248$, $N\le 128$)	2^{2m-F}	0 up to 183; 3, 7, 35 at 184, 185, 186	$m=187$
III ($F=376$, $N\le 194$)	2^{2m-F}	0 up to 279; 2, 3, 10, 49 at 280 to 283	$m=284$
V ($F=500$, $N\le 255$)	2^{2m-F}	0 up to 373; 2, 14, 67 at 374, 375, 376	$m=377$

Cost. At level I on KAT entry 0, medians of 201 iterations on the machine of §7.2: the two Weil ladders of the decompressor, of 128 and 127 steps, take 187.2 μs on E_{chl}; one Tate ladder of 187 steps takes about 152 μs, and 224.3 μs with the 61 doublings that bring the full-order basis to order 2^{187}. The fair comparison is the whole pairing stage. The decompressor first doubles the full-order bases 120 and 121 times (286.5 μs for both curves) and then runs two Weil ladders per curve (375.5 μs for both): 689.5 μs. The single-ladder stage doubles 61 times per curve and runs one ladder per curve: 453.1 μs, 34% less, which is 4% of a compressed verification (Table 4). At $m = F$ no doublings are needed and the ladder takes 181.8 μs per curve.

6.2 The metadata byte: distributions and non-eliminability

Table 2 gives the observed distributions of the three fields of the metadata byte over the 300 official KAT vectors. Table 3 gives the counts of the hint against the other two fields and against the parity of M_{00} at level I.

Table 2: Values of the metadata fields over 100 KAT vectors per level: counts of each value, smallest value first.

Field	**Level I**	**Level III**	**Level V**
n_{bt} $(0, 1, \ldots)$	77, 20, 1, 2	68, 25, 5, 1, 1	74, 17, 7, 2
r_{rsp} $(0, 1, \ldots)$	48, 24, 18, 4, 3, 0, 2, 1	39, 25, 18, 10, 6, 0, 1, 0, 1	47, 30, 14, 5, 2, 2
h $(0, 1)$	52, 48	50, 50	52, 48
chi-square of h, 1 d.o.f.	0.16	0.00	0.16

Table 3: Level I: the hint bit h against n_{bt}, against r_{rsp} and against the parity of M_{00}, counts over 100 vectors.

h	$n_{\mathrm{bt}} = 0, 1, 2, 3$	$r_{\mathrm{rsp}} = 0, \ldots, 7$	M_{00} even, odd
0	39, 11, 0, 2	28, 9, 9, 1, 2, 0, 2, 1	16, 36
1	38, 9, 1, 0	20, 15, 9, 3, 1, 0, 0, 0	12, 36

We considered three strategies for eliminating the hint bit h.

Signer retry. The signer could reject signatures whose hint does not match a fixed target, say $h = 0$, and sign again with fresh randomness. With $\Pr[h = 0] = 1/2$ the expected number of attempts is 2, and 95% success needs 5. Level I signing costs about 100 Mcycles in the reference implementation (Table 2 of the specification), so this is a slowdown of 2 to 5 times for one bit.

Algebraic elimination. Proposition 3.7 says what the pairings determine: $\det \mathbf{M}$ modulo 2^{N-1} from the logarithm at order 2^N. The remaining bit is the top bit of u in equation (1), a property of the signer's basis of E_{com} that the verifier never sees.

Deriving from other signature fields. The contingency counts above show no dependence of h on n_{bt}, r_{rsp} or the parity of M_{00} beyond what uniform sampling produces.

6.3 Inferring the metadata from the matrix

Reducing from 129 to 128 bytes would require eliminating the metadata byte. Corollary 3.6 gives $r_{\mathrm{rsp}} = v_2(\det \mathbf{M})$ once the matrix is known. But the matrix is known only after the determinant has been recovered at a precision that itself depends on n_{bt}, and n_{bt} is not a function of the matrix. We tested two readings of "the matrix determines its parameters" on the KAT vectors.

Consistency. For each vector we counted the pairs $(n_{\mathrm{bt}}, r_{\mathrm{rsp}})$ with $0 \le n_{\mathrm{bt}} \le 3$ and $0 \le r_{\mathrm{rsp}} \le 15$ for which all four entries lie below $2^{N(n_{\mathrm{bt}})}$, the bound of line 5 of Algorithm 4.9. No vector at any level has exactly one consistent pair, and the reason is structural. The bound depends on n_{bt} alone, so every value of r_{rsp} is consistent whenever n_{bt} is, and the count is always a multiple of 16. The weaker question, whether the matrix determines n_{bt}, has a partial answer. It does on 70, 61 and 67 of 100 vectors at the three levels, exactly those with $n_{\mathrm{bt}} = 0$ and some entry with bit $e_{\mathrm{rsp}} + 1$ set.

A leading one. A signer-side canonical form could place the parameters in the top bits of an entry, for instance by forcing M_{00} to start with a 1 followed by r_{rsp} and n_{bt}. The current signer does not do this. Bit $N-1$ of M_{00} is set on 49, 52 and 58 of 100 vectors at the three levels, and bit $N-1$ of at least one entry on 92, 89 and 88.

Either route needs the signer to produce a canonical matrix, which the v2.0 specification does not mandate; the round-3 specification does mandate a canonical sign (§8), but not a canonical top bit. The metadata byte stays.

7 Implementation and performance

7.1 Implementation

We implement a complete SQIsign v2.0 verifier in pure Rust. It has field arithmetic ($\mathbb{F}_p$, $\mathbb{F}_{p^2}$), Montgomery curve operations, the Weil pairing by the biextension cubical ladder, deterministic basis generation, $(2,2)$-isogeny chains in theta coordinates, and the verification flow with decompression. The implementation covers all three NIST security levels and all three signature formats (standard, expanded, compressed). The round-2 code measured in this paper is tag `v0.4.31` of the crate [Ray26b], the last round-2 release, with the format of §5; the round-3 format of §8 is in version 0.7.0.

Correctness is checked against the official NIST known-answer vectors of the reference at tag `nist-v2` [SQI25c], 100 per level. Every standard signature verifies. Every signature converts to the expanded format, which verifies after a byte round trip. Every signature compresses, decompresses to the byte-identical standard signature and verifies compressed. The counts are 300, 300 and 300 of 300.

The verifier crate is `no_std` with no heap allocation. All intermediate buffers are stack-allocated with sizes fixed at compile time by the parameter set. The crate contains no C code and no FFI calls.

7.2 Verification cost and its components

Table 4 is the cost of verification at level I on KAT entry 0 ($n_{\mathrm{bt}} = 0$, $r_{\mathrm{rsp}} = 1$). Each figure is the median or the minimum of 200 timed iterations after 5 warm-ups with `std::time::Instant`. Each component is timed in isolation on a copy of the state the real flow feeds it. The machine is an Intel Xeon at 2.80 GHz (Cascade Lake, a KVM guest with 32 virtual CPUs); the compiler is `rustc` 1.99.0. The release build has fat LTO and one codegen unit, supplied through the environment because the tag has no release profile of its own, and `target-cpu=native` from the tag's cargo configuration. Three runs agreed on every component median to within 0.2%.

Table 4: Level I verification cost in microseconds, KAT entry 0. The count column says how many times the standard path (S) and the compressed path (C) run the component.

Component	**S**	**C**	**median**	**S total**	**C total**
decode signature	1	1	0.2	0.2	0.2
range check of $\mathbf{M}$, curve checks	1	1	0.2	0.2	0.4
challenge derivation $E_{\mathrm{pk}} \to E_{\mathrm{chl}}$	1	2	960.6	960.6	1921.3
basis to hint on E_{chl} (hint check)	1	2	25.6	25.6	51.2
basis to hint on E_{aux} (hint check)	1	2	21.9	21.9	43.7
basis from hint, both curves	1	1	28.3	28.3	28.3
scaling the bases, both curves	1	2	286.1	286.1	571.9
Weil pairing on E_{chl}, order 2^N	0	1	188.4	0.0	188.4
Weil pairing on E_{aux}, order $2^{e'+2}$	0	1	187.3	0.0	187.3
inversions, point sums	0	1	8.5	0.0	8.5
discrete logarithm	0	1	90.3	0.0	90.3
entry recovery (inverse modulo 2^g)	0	1	0.4	0.0	0.4
basis change, apply $\mathbf{M}$	1	1	533.6	533.6	533.6
even part, $2^{r_{\mathrm{rsp}}}$-isogeny	1	1	52.5	52.5	52.5
$(2,2)$-isogeny chain	1	1	2163.2	2163.2	2163.2
final hash and compare	1	1	57.0	57.0	57.0
sum of component medians				4129.3	5898.1
full verification from bytes, median				4286.0	6005.7
full verification from bytes, minimum				4102.3	5902.8

The component sums fall short of the full measurements by 3.7% on the standard path and 1.8% on the compressed path, the cost of the glue between components. The compressed path costs 1720 μs more than the standard one at level I, 40%. At levels III and V the full medians are 11414 against 16629 μs and 23956 against 34258 μs, 46% and 43%.

The table says where that overhead is, and two costs follow from it. The separate path, which the round-2 code measured here implements, runs decompression as its own step that produces a standard signature and hands it to the standard verifier. It therefore repeats the challenge derivation, the two basis generations and the two basis scalings, 1294 μs of the 1769 μs of decompression-only components. It costs 1720 μs over the 4286 of a standard verification: 40% at level I, 46 and 43% at levels III and V. The integrated path shares the challenge derivation and the basis work with verification and adds only the two Weil pairings, the inversions and point sums, the discrete logarithm and the entry recovery. From the table that is $375.7 + 8.5 + 90.3 + 0.4 = 474.9$ μs, 11% of a standard verification. No extra basis scaling is needed, because the verifier already scales both bases to the orders the pairings use. The round-3 implementation of §8 is integrated and measures 13 to 18%. The pairings alone are 376 μs, 6% of the separate path's compressed verification.

Without `target-cpu=native` the full medians are 4394 and 6256 μs at level I; with the tag's own default profile (no cross-crate LTO) they are 4504 and 6229. The absolute figures are those of this machine and build and are not comparable to figures on other hardware.

8 Round 3

The round-3 specification [SQI26] of 1 September 2026 changes the primes to $3 \cdot 2^{324} - 1$, $27 \cdot 2^{500} - 1$ and $17 \cdot 2^{664} - 1$ (its Chapter 5), after the attack of Wesolowski [Wes26]. It also simplifies the response. The quaternion response has odd norm (Algorithm 3.18), so there is no even part and no r_{rsp}. The challenge has length λ and there is no backtracking count. The response bound

is $e_{\mathrm{rsp}} = \lceil \log_2 p/2 \rceil + \lambda/4 + 1 = 196,302,400$ (Table 5). Both deterministic bases are scaled to order $2^{e_{\mathrm{rsp}}+2}$ (Algorithm 3.24, lines 9 and 10). The matrix has entries in $[0, 2^{e_{\mathrm{rsp}}+2})$ and is normalized to the lexicographically smaller of $\mathbf{M}$ and $-\mathbf{M}$ (Algorithm 3.10, applied at line 19 of Algorithm 3.23 and checked at line 4 of Algorithm 3.24); this removes the sign freedom described in [Ray26c]. The signature is $(\mathsf{chl}, E_{\mathrm{aux}}, \mathbf{M}_{\mathrm{chl}}, \mathsf{hint}_{\mathrm{aux}}, \mathsf{hint}_{\mathrm{chl}})$. It is encoded in $2 \cdot \texttt{FP_ENCODED_BYTES} + 4 \cdot \texttt{RESPONSE_BYTES} + \texttt{CHALLENGE_BYTES} + 2$ bytes with $\texttt{RESPONSE_BYTES} = \lceil (e_{\mathrm{rsp}} + 2)/8 \rceil = 25, 38, 51$ (Chapter 6, Table 11): 200, 306 and 406 bytes. The full matrix is transmitted. The specification does not compress it, as De Feo's replies of July and October 2026 anticipated (§1).

Theorem 3.2 applies with $n_{\mathrm{bt}} = r_{\mathrm{rsp}} = 0$: $N = e_{\mathrm{rsp}} + 2$, $g = e_{\mathrm{rsp}}$, both pairings are taken at order 2^N, the determinant is odd, and a pivot among M_{00} and M_{01} always exists. Our round-3 format, in version 0.7 of the crate [Ray26b], transmits three entries of $\lceil e_{\mathrm{rsp}}/8 \rceil$ bytes (the low e_{rsp} bits), one byte holding bit e_{rsp} of those three entries, the challenge, and the two hints. That is $2 \cdot \texttt{FP_ENCODED_BYTES} + 3\lceil e_{\mathrm{rsp}}/8 \rceil + \lambda/8 + 3 = 176,269,353$ bytes. The fourth entry is recovered modulo $2^{e_{\mathrm{rsp}}+1}$, so its bit e_{rsp} is computed, not sent. Three bits ride along because the verifier's chain consumes the points P and Q themselves, not only the kernel they generate, and its acceptance depends on bit e_{rsp} of every entry. Verification does not depend on bit $e_{\mathrm{rsp}} + 1$ of the entries, for two reasons in the specification. Adding $2^{e_{\mathrm{rsp}}+1}$ to an entry adds a point of order 2 to P or Q, and the chain of Algorithm 3.20 uses the two points only through their multiples: the kernel $\langle [4]P, [4]Q \rangle$ (Algorithm 3.19) and the 8-torsion points above it that the gluing step consumes (Section 2.4.1 of the v2.0 specification), none of which a 2-torsion summand changes. The sign check of Algorithm 3.24, line 4, reads that bit only on the first entry that is neither 0 nor $2^{e_{\mathrm{rsp}}+1}$, and the signer's own normalization (Algorithm 3.10) leaves it zero there. The decompressor therefore leaves bit $e_{\mathrm{rsp}} + 1$ zero in every entry and runs the standard verification, sign check included, on the result. The round-2 format carries the dropped entry's top bit in its hint so that the round trip reproduces the signer's bytes (§4.3); the round-3 format does not need to. The two hint bytes are kept; recomputing them as in §5.2 would give 174, 267 and 351 bytes and is not done in the crate.

Table 5 is the measurement on the 300 round-3 KAT vectors of the reference [SQI26b] at tag `nist-v3`. Every vector compresses, verifies compressed, and decompresses to a signature the standard verifier accepts. The decompressed matrix equals the signer's in 14, 14 and 13 of 100 cases and differs only in bit $e_{\mathrm{rsp}} + 1$ of some entry on the other 86, 86 and 87, all of which verify.

Table 5: The round-3 format on the 300 round-3 KAT vectors. Pivot path: vectors with M_{00} even, where M_{11} is transmitted instead of M_{10}.

Level	Standard	Compressed	Saved	Verified	Pivot	Exact
I	200 B	176 B	24 B (12.0%)	100/100	32/100	14/100
III	306 B	269 B	37 B (12.1%)	100/100	22/100	14/100
V	406 B	353 B	53 B (13.1%)	100/100	37/100	13/100

The recovery costs two Weil pairings at order $2^{e_{\mathrm{rsp}}+2}$ and one discrete logarithm on top of a verification. On the crate's benchmark machine that is 2.2, 5.2 and 11.0 Mcycles, 18, 17 and 13% of a verification under a prepared key [Ray26b].

9 Conclusion

We have proved, from the definitions of the SQIsign v2.0 specification, that $\omega^{\det \mathbf{M}} \eta$ is a primitive fourth root of unity, for ω and η the Weil pairings of the transmitted challenge and auxiliary bases. This gives $\det \mathbf{M}$ modulo $2^{e_{\mathrm{rsp}} - n_{\mathrm{bt}} + 1}$, one bit short of the width the matrix is carried at. With a

one-bit hint, an adaptive pivot, recomputed basis hints and a packed metadata byte the level I signature shrinks from 148 to 129 bytes, checked on the 300 NIST KAT vectors. The reduced Tate pairing cannot replace the Weil pairing here: on the transmitted basis its values have order at most 2^{2N-F}, and a symmetric pairing does not see the determinant. On the round-3 parameters the same recovery gives 176, 269 and 353 bytes against 200, 306 and 406.

Why the bytes matter. SQIsign's value is its combined size, 65-byte public keys and 148-byte signatures at level I in round 2 (Table 1 of the specification). A LoRaWAN uplink at spreading factor 7 carries at most 242 bytes of payload, so a 129-byte signature leaves 113 bytes for the message where 148 leaves 94. A DNS response above 512 bytes falls back to TCP and costs a round trip, and every byte of signature in a DNSSEC answer comes out of that budget. In a TLS 1.3 handshake with a three-signature chain the saving is 57 bytes; in a transparency log or a ledger it is 19 bytes per entry, permanently.

Future work. A pairing-free recovery along the lines of Pereira and Barreto [PB21] would need the Pohlig-Hellman step on the verifier's curves rather than on E_0; whether that can be done at useful cost is open. Reaching 128 bytes needs a signer-side canonical matrix (§6.3). The dimension-4 compact format of the same crate, 108 bytes at the round-2 parameters and 142 at round 3, is the subject of a separate report [Ray26b].

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